NARI V. MILLER

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Education

2024 Ph.D. in Geological Sciences
Arizona State University, Tempe, AZ
Dissertation Title: *Linking Process and Form in Carbonate Rock through Cosmogenic 36-Chlorine Erosion Rates, Regolith Mass Balance and Fluvial and Hillslope Topography.*

2012 B.A. in Geosciences (with Honors) and Chemistry
Williams College, Williamstown, MA

Select Classes and Research Skills

2016 Numerical Methods, ASU
Learned and implemented numerical methods for filtering and modeling.

2016 Crustal Deformation, ASU
Worked with data collected via InSAR to detect crustal flexure information.

2017 Remote Sensing, ASU
Worked with LANDSAT and other satellite raster products to map geologic features on Earth as an analog to planetary research.

2017 Advanced Geomorphology, ASU
Geospatial analysis of topography, river networks, hillslope gradients and fault systems using Python, MATLAB and ArcGIS Desktop.

2017-2023 Geomorphology Research, ASU
I managed related field, lab, and analysis projects. I constructed geospatial databases that tied sample origin with geochemical composition (lab results). I became an intermediate user of python, incorporating scripts to analyse and visualise cross-platform data, such as TRMM, geologic maps, and GPS-derived uplift rates. I used Pandas, raster analysis, shapefile construction and modification, matplotlib, among other techniques.

Teaching Experience

Summer 2025, 2026 **Water Planet**, as instructor of record. *A general-education science course that teaches quantitative thinking in the context of the energy budget of Earth.* Arizona State University, Tempe, AZ.

S20, S22 **Geochemistry**, as instructor of record. *Upper-level course for undergraduate Geology Majors.* California State University, Stanislaus, Turlock, CA
I incorporated case studies of geochemistry research to connect classroom topics with relevant issues. In writing their final project, students had to use campus library resources, create their own conceptual diagram and map of the data locations in context.

F21 **Earthquakes & Volcanoes**, as instructor of record. *General-education science course.* California State University, Stanislaus, Turlock, CA

I combined engaging videos and discussions of earth hazards in historical context with cultural and scientific perspectives.

2016 – 2019 **Field Geology II** in Tonto National Forest, AZ, and San Juan National Forest, CO, as teaching assistant during the 3-week summer course. Primary instructors: Tom Sharp and Arjun Heimsath. Arizona State University.

Taught field observation and hypothesis testing, digital mapping, and note-taking to students in the field. Tutored students on interpreting and synthesizing their observations into reports.

Organized logistics for the 22-student, three-week course, including food, vehicles, office materials.

F17 **Earth's Critical Zone**, as teaching assistant. Arizona State University, Tempe, AZ. Held office hours and graded homework.

F14 - S16 **Historical, Physical, and Introductory Geology**, as teaching assistant. Arizona State University, Tempe, AZ. As the TA, I taught the lab sections.

Introductory hands-on geology labs for non-majors. Included short campus field trips to see local unconformities and fossils. Coordinated introductory geology graduate TA's as Head TA (F15 - S16)

Field Research Experience

2018 Collaborated with University of Washington archaeology group (PI: Marcos Llobera) in the Son Servera Valley, Mallorca, Spain, and produced a geomorphic map to aid archeology field campaign planning.

2015 - 2017 Conducted fieldwork near Valencia, SP, for dissertation: collected soil and bedrock samples for erosion measurement using cosmogenic ^{36}Chlorine and *meteoric* ^{10}Beryllium.

2014 Collected shallow seismic measurements with a group to characterize material properties of different lithologic layers in the Grand Canyon (Nov 2014) and on a river trip in Desolation Canyon (April 2015).

2012 Albion College Field Camp, Wyoming and South Dakota

Lab Research Experience

2016 - 2019 Soil and rock processing & pulverization for major, trace, and cosmogenic radionuclide element analysis in AZ.

2018 Conducted standard methods of isotope dilution to measure ^{36}Chlorine in carbonate samples at the University of Köln, Germany.

2017 Extracted meteoric ^{10}Be from soils in University of Vermont's CosmoLab.

2013 National Association of Geoscience Teachers Intern, with Water, Energy, and Biogeochemical Budgets Project, USGS, Boulder, CO.

Measured dissolved oxygen, turbidity, ions from long-term monitored sites. Collected water samples in Nome Creek Watershed, AK.

Professional Development

2021 CIENCIA Summer Institute, Summer semester, CSU Stanislaus.
The Collaboration for Inclusive and Engaging Curriculum, Instruction, and Achievement brings STEM faculty together to learn and discuss best practices for teaching students from diverse backgrounds.

2020 & 2021 Quality Learning and Teaching Program, Summer semesters, CSU Stanislaus.
Learned best practices for clear and effective online teaching.

Publications, Conference Posters, and Presentations

In Prep **Miller**, N, Heimsath, A., Bierman, P., Corbett, L. and Barton, M. "Partitioning Chemical and Physical Erosion of Regolith and Carbonate Bedrock in SE Spain and Arizona, USA, Using a Mass Balance Model."

2024 **Miller**, N., Heimsath, A., Bierman, P., Corbett, L., and Barton, M. Dust input to regolith and chemical erosion of carbonate hillslopes: a mass balance approach. Abstract 104-5. In Session "T39. What's the Cosmognosis? Recent Advances in Understanding Earth and Planetary Processes with Cosmogenic Nuclides (Posters)." Anaheim, CA, 23 September.

2016 Cook, M, Ravelo, A, Mix, A, Nesbitt, I and **Miller**, N,"Tracing subarctic Pacific water masses with benthic foraminiferal stable isotopes during the LGM and late Pleistocene," Deep Sea Research Part II: Topical Studies in Oceanography, March, Vol 125, pp 84-95. DOI: 10.1016/j.dsr2.2016.02.006.

2014 Cook, M., Ravelo, A., Mix, A., Nesbitt, I., **Miller**, N., *Tracing Bering Sea Circulation With Benthic Foraminiferal Stable Isotopes During the Pleistocene*, Abstract PP23D-08 presented at AGU Fall Meeting, San Francisco, CA, 15-19 December.

2012 **Miller**, N. and Cook, M., *Evidence for elevated methane flux in laminated Bering Sea sediments from the penultimate glaciation*, Abstract PP13B-2105 presented at AGU Fall Meeting, San Francisco, CA, 3-7 December.

Grants and Awards

2018 Free Seed Sample Analyses at PRIME Lab, Purdue University.
Awarded about $11,700 value of cosmogenic ^{36}Chlorine sample preparation and AMS measurements for a study of mechanical and chemical erosion and multi-scale sediment production in limestone terrain.

2016 Woodside Grant Recipient
Project Coordinator, $1,500. Initiated and mentored an undergrad in a research project analyzing heavy metals in local community gardens.

2012 The David Major Prize for Excellence in Geosciences, Williams College.

Service

2025-present	Geomorphica Copyeditor I helped develop the copyediting guidelines for the journal Geomorphica, and I copyedit accepted papers, which is free to read and does not charge article processing fees.
2025 & 2026	Proposal Reviewer I reviewed student grant proposals for the Quaternary Geology and Geomorphology Division of the Geological Society of America.
2025 & 2026	Santa Barbara County Science and Engineering Fair Activity Coordinator. I organized lab tours and activities at University of California, Santa Barbara, for 40 high school science fair students.
2024	Quaternary Geology and Geomorphology Division of Geological Society of America conference volunteer.
2015 - 2018	School of Earth and Space Exploration Colloquium Search Committee, Arizona State University.

Technical Expertise

Software Proficient in LaTeX, Excel, Inkscape, ArcGIS

Data Analysis and Visualization

Python (Proficient): Pandas, Matplotlib, GeoPandas, Streamlit, Landlab.
I created an interactive Mass Balance Model to accompany my 2024 GSA Poster. `https://carbonate-regolith.streamlit.app/`
MATLAB (Proficient): TopoToolbox (topographic analysis), CRONUScalc

References

Professor Arjun Heimsath	aheimsat@asu.edu	(480) 965-5585
Professor Kelin Whipple	kxw@asu.edu	(480) 965-9508
Professor Tom Sharp	tom.sharp@asu.edu	(480) 965-3071
Professor Rob Rogers	rrogers1@csustan.edu	(209) 664-6691